

SEA LION ISLAND, Falkland Islands (Malvinas)

WMO: 888970

Lat: 52.426S

Lon: 59.076W

Elev: 49

StdP: 14.67

Time Zone: -3.00 (W03)

Period: 91-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	29.3	31.2	20.4	15.3	32.3	22.6	17.1	33.3	39.1	37.4	35.2	38.1	17.3	190	0.932

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.1	60.2	54.4	58.1	53.1	56.3	52.0	55.7	58.5	54.1	57.0	52.8	55.4	13.8	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
54.3	62.7	56.7	52.7	59.2	55.3	51.2	55.9	53.9	23.8	58.8	22.8	57.0	22.0	55.6	61.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.3	32.2	29.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.8	49.5	49.5	48.7	44.7	41.7	38.7	38.2	38.2	40.5	43.0	45.7	47.9
	DBStd	5.48	3.29	3.82	3.79	3.82	3.35	2.96	3.32	3.36	3.24	3.48	3.51	3.93
	HDD50	2385	49	50	71	162	257	338	366	365	284	218	136	89
	HDD65	7728	480	435	506	609	722	788	831	830	734	682	580	531
	CDD50	133	34	36	30	3	0	0	0	0	0	1	6	23
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	16.2	16.2	14.8	15.2	16.9	16.0	16.2	16.8	17.2	15.6	16.0	17.3	15.6
Precipitation	PrecAvg	31.5	2.7	2.5	3.1	3.0	2.9	2.7	2.6	2.6	2.3	2.4	2.2	2.5
	PrecMax	42.3	3.8	3.7	4.8	5.0	4.5	5.4	5.4	4.7	5.5	6.4	3.7	4.8
	PrecMin	25.4	1.7	1.5	1.7	1.3	1.5	1.2	1.3	1.5	1.1	1.3	1.3	1.3
	PrecStd	3.9	0.6	0.6	0.7	0.9	0.7	0.9	0.9	0.7	0.9	0.9	0.5	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.2	62.4	60.9	54.9	51.2	46.3	47.4	46.4	51.2	55.6	59.3	62.7
		MCWB	55.8	57.5	55.4	51.2	47.7	44.3	44.4	43.0	46.3	49.7	51.6	55.1
	2%	DB	59.1	59.5	57.9	52.9	49.1	45.0	45.5	44.9	48.7	52.5	55.7	58.9
		MCWB	54.1	54.8	53.7	49.8	46.8	43.5	43.0	42.1	44.7	47.9	50.1	52.5
	5%	DB	57.0	57.1	56.0	51.3	47.8	43.9	44.0	43.7	46.6	50.2	53.2	56.1
		MCWB	52.4	53.0	52.5	48.8	46.0	42.5	41.8	41.3	43.4	46.1	48.4	50.9
	10%	DB	54.8	55.0	54.2	50.0	46.6	42.9	42.8	42.6	44.9	48.4	51.2	53.7
		MCWB	51.0	51.7	51.3	48.0	44.9	41.5	40.9	40.5	42.4	44.8	47.1	49.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.9	58.3	56.8	52.2	48.7	45.0	44.8	43.6	46.7	51.2	53.1	56.4
		MCDB	60.6	62.4	59.2	53.8	50.1	45.5	46.7	45.2	50.5	54.4	57.5	60.9
	2%	WB	55.1	55.6	54.7	50.6	47.5	43.8	43.3	42.5	44.9	48.6	50.8	53.7
		MCDB	58.0	58.1	56.7	52.0	48.6	44.4	44.8	44.2	48.0	51.7	55.0	57.9
	5%	WB	53.2	53.8	53.0	49.3	46.3	42.8	42.1	41.6	43.8	46.5	48.9	51.6
		MCDB	56.2	56.3	55.2	50.8	47.5	43.7	43.5	43.2	46.3	49.5	52.4	55.1
	10%	WB	51.5	52.3	51.7	48.1	45.1	41.7	41.1	40.8	42.7	45.0	47.5	49.9
		MCDB	53.9	54.5	53.7	49.8	46.4	42.7	42.5	42.3	44.5	47.8	50.6	53.2
Mean Daily Temperature Range	5% DB	MDBR	8.9	8.1	7.2	6.4	5.1	4.7	4.9	5.5	7.3	8.5	9.2	9.7
		MCDBR	11.8	11.0	9.8	8.0	6.0	5.2	5.8	6.4	9.5	10.6	12.7	13.6
		MCWBR	8.0	8.0	7.2	6.5	5.4	5.3	5.4	5.4	6.9	7.1	8.4	8.8
	5% WB	MCDBR	11.3	11.1	9.6	7.6	5.8	5.2	5.3	6.2	9.2	10.2	11.8	12.7
		MCWBR	8.2	8.5	7.7	6.6	5.6	5.5	5.0	5.5	7.0	7.2	8.6	9.0
Clear-Sky Solar Irradiance	taub	0.342	0.335	0.327	0.324	0.307	0.300	0.306	0.318	0.322	0.325	0.330	0.339	
	taud	2.505	2.532	2.539	2.475	2.374	2.285	2.336	2.425	2.491	2.499	2.507	2.499	
	Ebn at Noon	305	296	279	245	207	182	202	241	275	296	308	309	
	Edh at Noon	33	30	26	22	18	16	18	23	27	31	33	34	
All-Sky Solar Radiation	RadAvg	1891	1492	1020	557	279	173	210	434	834	1389	1834	2096	
	RadStd	78	101	78	43	16	11	14	33	66	81	128	176	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	+1.06	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page